

The Companion That Notices

An AI-initiated family safety sentinel: the journey dialogue and escalation grammar, pack-watch silence-as-alarm over a serverless kin topology, duress-preserving cancellation, a verified-responder proximity mesh, break-glass pre-consented data release, privacy-by-distillation at an expiring forensic window, and the anonymous courier with returned witness — specified together so that a safety product is structurally hard to turn into a surveillance product.

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Note. This defensive publication specifies the mechanism family of **B-SafetySM**, the safety app of the HeartBank® architecture, and of **B-SidekickSM**, the companion that lives in it, together with the mesh that answers its bark (**B-PackSM**) and the courier mechanisms of its physical sibling (**B-DogTM**, the flying bot that delivers, and protects only as a witness; §13). It is published at the design stage, ahead of any build, because personal safety is an actively iterated product space — a platform owner could absorb any one of these mechanisms in a single operating-system release — and because the combination specified here, in particular the pack-watch inversion (claim 2), the break-glass advance directive (claim 5), and the distillation boundary (claim 6), is what this publication discloses. Companion works: *The Thank-All-Nearby Primitive* (the proximity substrate the responder mesh reuses), *Proof of Coordinate* and *B-PoHSM as Humanity Layer* (the identity primitives the mesh and the courier lean on), *The Time-Locked Gift Tag* (the per-field reveal and time-lock grammar the break-glass release inherits), *Verified-Human Anonymous Local Gratitude Transfer* (the accountable-anonymity pattern the courier extends), and *The Gift Operation* (the circulation grammar in which a safety floor is the precondition of the stranger-gift).

Terminology note (2026-09-23). The companion family's names were reassigned on 15 September 2026, and this revision adopts them. As first published and prior-art-snapshotted (4 July 2026), this paper called the companion on its holder's phone the "digital **B-DogSM**" and reserved **B-SidekickTM** for an aerial courier — so the old names now mean the *opposite* of what they meant here: in an earlier version a "B-Sidekick" is an aircraft; in this one it is the companion on the phone. The current names are: **B-SafetySM** — the app; **B-SidekickSM** — the holder's own companion inside the app, the one that barks (formerly "digital B-DogSM"); **B-PackSM** — the mesh that answers a bark; **B-DogTM** — the physical flying bot, which delivers and protects (the aerial role earlier reserved as B-SidekickTM); it protects only as a witness that flies to a bark, never on patrol and never as the rescue, a role specified in *The Quest*

That Mints Nothing (claim 8) rather than here. Unlike this corpus's earlier terminology note on the Re-Tip Jar, which kept the original label for citation continuity, this revision changes the names in the text, because an old name left in place would now name a different thing; a reader citing an earlier version should translate by this table. At that revision the mechanisms specified in this document and its seven claims were unchanged; only the product labels moved (the rewording of claims 1, 4 and 5 on 1 October 2026 is noted after the claims). One rule travels with **B-PackSM**: *pack* means the holder's own people. The kin pack of §5 and the nearby verified responders of §6 are opted into separately and presented visibly differently, and the name never lends a stranger the trust of family (§6).

Preamble

*This specification is offered to the commons in the spirit of **dāna** — the gift that asks nothing back. May the alarm it specifies never need to sound; may every companion it describes die of old age having never once bitten; and may the record of where a person walked belong, always and only, to the person who walked.*

Every personal-safety product eventually faces one question, and almost none of them survive it: *what is the difference between watching over someone and watching someone?* The two activities use the same sensors, the same data, the same map with the same moving dot. The difference is not technical; it is grammatical — it lives in who initiates, who holds the record, who can open it, and what the record even is. The incumbent family-location products answer the question badly and profitably: they retain weeks of movement history in the vendor's cloud, they sell access to that history as their premium tier, and — in the most documented case — they sold the underlying data itself to brokers. The result is a category in which, by the published findings of the intimate-partner-surveillance research community, most of the apps relevant to partner surveillance are "dual-use": legitimate child-safety and anti-theft tools, easily repurposed (§2.3).

This paper specifies a safety companion built so that the question has a structural answer rather than a policy answer. The companion initiates, so that protection does not depend on a frightened person remembering to arm an app. The alarm survives the destruction of the phone that would have raised it, because the watch is kept by the family's devices collectively, not by the traveler's device alone — and *silence itself is the alarm*. The forensic record that a rescue genuinely needs is mortal by construction: a short rolling window that expires on schedule, releasable only under an advance directive its owner wrote in calm daylight, triggered only by the alarm event, never by a relative's request and never by a payment. What persists past the window is not the track but a distillation of it — a diary of places, written at day's end on the owner's own device, erasable page by page, owned absolutely by the person who walked. And the guardianship asymmetry is stated as an invariant rather than a setting: *a guardian receives the alarm, never the archive*.

The specification closes with a mechanism from the same companion family that is not about alarms at all: a third-party gift courier that offers something hand-delivery cannot — a gift given anonymously *whose gratitude nevertheless finds its way back to the giver*, carried home by the conveyance that delivered it. Anonymity and witness, simultaneously.

We publish the family together because the mechanisms share one root: a companion agent whose loyalty runs to its holder and to no one else — not to the institution that made it, not to the relative who pays, not to the data broker, not to the court that subpoenas a cloud that was never given anything to hold.

Prior-Art and Non-Assertion Statement

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This document constitutes a defensive publication establishing **prior art as of 4 July 2026** for the combination of mechanisms described herein. The following mechanisms are disclosed here, each as specified and in the combination specified, and any subsequent patent application claiming them should be considered filed against this disclosure:

1. **The AI-initiated journey sentinel** — a safety agent that *initiates* the protective dialogue rather than waiting to be armed: on detecting that its holder has begun moving, the agent opens a lightweight query ("where are you headed?"), derives a reasonable route and arrival-time envelope from the answer, monitors the live journey against that envelope entirely on-device, and escalates through a graded alert ladder on wide divergence or non-arrival — inverting the manually-armed check-in pattern (in which the person must remember to start a session per trip) so that protection no longer depends on remembering to arm it; a dismissed or ignored prompt still starts no monitored journey (§4, §12).
2. **Pack-watch: silence-as-alarm over a serverless kin topology** — the journey envelope (destination class, expected route corridor, deadline) replicated end-to-end-encrypted from the traveler's device to a pre-designated kin set (the "pack"), whose devices *independently* run the same deadline timer and expect a periodic authenticated heartbeat from the traveler's agent; expiry of the deadline *or cessation of the heartbeat* raises the alarm on the pack's devices **without any action by, or survival of, the traveler's device** — so that destroying, disabling, or jamming the traveler's phone is itself the alarm, the dead-man default of industrial safety engineering transposed to a peer family topology with no server in the loop.
3. **Duress-preserving cancellation with graded escalation** — a cancellation control that is visually and behaviorally indistinguishable from an ordinary cancel to an observer physically present, but which (when invoked via the holder's pre-set duress variant) suppresses the local alarm while still transmitting it to the pack; combined with an escalation ladder (companion-to-holder query first, kin alert after a response window, context-adaptive timing by travel mode) calibrated against alarm fatigue.
4. **The verified-responder proximity SOS mesh** — an opt-in mesh in which nearby holders of the same companion may volunteer to receive a stranger's SOS, gated by personhood verification of the responder, with responder identity accountability-logged at opt-in and at response, victim location released to

responders only at coarse resolution until a responder is vetted or accepted, and the full alarm dispatched simultaneously to the victim's kin set and (where integrated) emergency services — so that the mesh adds bystander capability while any responder, including a bad actor who passes verification, receives only a coarse position until a specific responder is accepted, and every response is attributable (§6, §11).

5. **Break-glass pre-consented release — the advance directive for location data** — the holder, in advance and in calm conditions, authorizes the release of a *bounded recent window* of their movement record (e.g., the trailing seventy-two hours to seven days) to a named kin set, with the release *triggerable only by the alarm event itself* (loud bark, SOS, or sustained heartbeat loss); the release is event-gated and consent-predates-crisis: it cannot be invoked by a relative's request, purchased at any price, or compelled by the institution, because outside the triggering event no single party holds both the encrypted window and a path to its key, and the institution holds neither (§7; for the heartbeat-loss trigger, see the residual in §11).
6. **Privacy-by-distillation at an expiring forensic window** — a two-tier movement-data covenant in which (i) the raw, forensic-grade track exists only within a short rolling window, replicated solely as the encrypted break-glass envelope, and *expires by construction* on schedule; and (ii) before expiry, an on-device process distills each day into a low-resolution semantic *place-diary* entry (named places and durations, never a continuous coordinate trail), such that long-term record and surveillance-grade record are never the same object; with the diary governed as autobiography — erasable page by page by its owner, subject to an owner-configured sensitive-place never-write list, distinct from any integrity-bearing append-only ledger in the same product — and bound into a permanent artifact only by the owner's explicit year-end act.
7. **The anonymous courier with returned witness** — third-party gift conveyance (robotic or human courier) in which the sender's identity is withheld from the recipient while remaining internally signed and accountable (personhood-verified at the system layer), the conveyance carries a *return channel* by which the recipient's freely offered thanks travels back to the anonymous sender — witness without unmasking — and sender anonymity is subordinate to recipient protections: an anonymous conveyance can never reach a recipient who has blocked the sender, and recipients may refuse anonymous conveyance categorically.

The component lineages — manually-armed check-in and ETA-sharing applications; dead-man's switches, watchdog timers, and heartbeat monitoring in industrial and distributed-systems engineering; end-to-end encryption and encrypted peer device meshes; personhood verification; crash- and fall-detection auto-alerts; inactivity-triggered data bequest; advance directives and Ulysses contracts; data-minimization and on-device semantic location processing; anonymous gifting conventions — are prior art and are cited generously in §2 and §14. The *synthesis*, and in particular the pack-watch inversion (claim 2), the advance-directive release (claim 5), the distillation covenant (claim 6), and the returned-witness courier (claim 7), are the contribution. These claims were drafted before this institution adopted a pre-registered census of the world's prior art for its defensive publications; no such census has been run against them, they rest on the review in §2 and §14, and none is asserted to be new — only to be disclosed, in the form given, as of the date above.

Cases that cut against, added at the 1 October 2026 revision. A second desk review (not a pre-registered census) found three ancestors closer than the first review recorded, and each narrows a claim. Apple's Check

In already runs silence-as-alarm over end-to-end-encrypted data, with the vendor's servers holding the heartbeat clock and the decryption key (§2.1): claim 2 survives only in its serverless kin form — the watchdog held independently on several family devices, no server in the detection or release path — and claim 5 only as a trailing window held by kin as ciphertext under a kin-quorum key path, with no server holding a key. The Companion app (2015) already ran the divergence, query and short-cancel pattern that claim 1 monitors with and claim 3 grades, started manually per trip (§2.1). Lone-worker monitoring already treats an uncanceled timer, including one set before a no-signal area, as an alarm at a staffed centre (§2.2). Claims 1, 4 and 5 were reworded at the same revision to state their guards at the width that §4, §6, §7 and §11 support; no matter was added to them.

Trademark rights on specific marks — **B-SafetySM**, **B-SidekickSM**, **B-PackSM**, **B-DogTM**, **B-QuestSM**, **B-HauntSM**, **B-CatTM**, **B-DatingSM**, **B-LetterSM**, **B-Yearbook**, **3B-DiamondTM**, **HeartBank®**, **Miss AquariusSM**, **Proof of HumanitySM**, **PoHSM**, **Proof of CoordinateSM**, **PoCSM**, the B-heart logo (**B-EmblemTM**) — are separately and explicitly reserved. The *mechanism* is dedicated to the commons; the *marks* are not.

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Abstract

We specify a personal-safety mobile application in which an on-device software agent starts journey monitoring itself, a dead-man's-switch alarm runs independently on family members' devices with no server in the loop, a duress code cancels visibly while still alerting, opt-in verified volunteers can respond to an SOS, a short encrypted location history is released only on an alarm event, long-term location data is minimised on the device, and an anonymous gift delivery carries a reply channel back to the sender — published as **B-SafetySM**, a family safety app, with the companion that lives in it, **B-SidekickSM**. The governing constraint is that a safety product must be structurally incapable of functioning as a surveillance product; §11 names where the design falls short of it. Six mechanisms carry the constraint. **(1) The journey sentinel initiates:** when the companion notices its holder in motion it asks for the destination, computes a route-and-arrival envelope, and monitors the live journey on-device — inverting the manually-armed check-in whose canonical failure is that no one arms it on the night it matters. **(2) Pack-watch makes silence the alarm:** the envelope replicates end-to-end-encrypted to the family's devices, which run the deadline independently and expect the traveler's heartbeat; a destroyed, dead, or jammed phone therefore raises the alarm rather than suppressing it, with no server in the loop. **(3) Cancellation is duress-preserving** and escalation is graded against alarm fatigue. **(4) A proximity SOS mesh** adds verified, accountability-logged volunteer responders under coarse-until-vetted location release, reusing the architecture's anonymous-proximity primitive; a bad actor who passes verification still receives only a coarse position until accepted, and is identifiable. **(5) Break-glass release is an advance directive for data:** a bounded trailing window of

movement record, pre-consented by its owner in calm conditions, releasable to named kin only by the alarm event — never by request, payment, or institutional decision. **(6) Privacy-by-distillation separates the two records forever:** the forensic track is mortal (a rolling window that expires by construction) while the persistent record is a semantic place-diary distilled nightly on-device — autobiography, not tracking log — erasable by its owner, bound into a permanent family artifact only by an explicit year-end act. A seventh mechanism extends the companion family to conveyance: **the anonymous courier with returned witness**, in which third-party delivery lets a whole gift arrive with its sender withheld, while a return channel in the conveyance itself carries the recipient's thanks back to the unnamed giver — anonymity and witness simultaneously — under absolute recipient protections (blocks are never defeated; anonymous conveyance is refusable categorically). Constitutional invariants bind the whole: the safety capability is free universally ("rich or poor"), access to another person's past is never a product at any price or any tier, and a guardian receives the alarm but never the archive — the asymmetry that keeps the product safe *inside* the family, which is where the incumbent category demonstrably failed. Everything is computed on the holder's device, and nothing is stored where the institution can read it; the companion's loyalty runs to its holder alone. The design is published before it is built ($n = 0$) and offered defensively to the commons under CC0.

Keywords: personal safety application, safety check-in, journey monitoring, estimated time of arrival, geofence, dead man's switch, watchdog timer, heartbeat monitoring, emergency alert, SOS, duress code, duress PIN, family location sharing, end-to-end encryption, peer-to-peer, store-and-forward relay, threshold cryptography, emergency access, conditional data release, location history, data retention, data minimization, on-device processing, semantic location, location privacy, intimate partner surveillance, stalkerware, dual-use apps, volunteer responder network, proximity alerting, location obfuscation, proof of personhood, anonymous gift delivery, delivery drone, recipient blocking, defensive publication.

Connection to the unified mission frame. This specification serves HeartBank's canonical mission — a reciprocity infrastructure in which every human being is treated as uniquely different and equally necessary — at the mission's precondition layer. The architecture's telos includes a world in which "stranger danger" is no longer a reason to withhold kindness from a stranger; a gift economy between strangers presupposes streets on which strangers are safe. The companion specified here is that presupposition built: its success metric is a lifetime of silence, and the institution that operates it is permanently denied the one asset — retained human movement history — that would make it valuable to everyone except the people it protects.

Terms

Coined names used in this paper and the standard terms an examiner would search for them.

Term used here	Standard technical term
B-Safety SM	personal-safety mobile application
B-Sidekick SM ; the companion	on-device software agent (safety assistant) bound to one user
holder	user; the protected person
soft bark	agent-initiated check-in prompt (push notification)
insistent bark	escalated prompt with a visible cancel window
loud bark	emergency alert to designated contacts
journey sentinel	automatic trip detection with a destination query and arrival monitoring
journey envelope	expected route corridor and arrival-time window (corridor geofence and deadline)
pack; kin set	designated emergency contacts drawn from the family group
pack-watch; silence-as-alarm	dead man's switch / watchdog timer run on the contacts' devices; heartbeat-loss alarm
heartbeat	periodic signed keep-alive message carrying no location
duress cancel	duress code (duress PIN): a covert distress signal disguised as a normal cancel
B-Pack SM ; responder mesh	opt-in volunteer responder network with proximity-based SOS dispatch
coarse-until-vetted	location precision reduction until a responder is accepted
PoH SM	proof of personhood (identity verification of a unique human)
PoC (Proof of Coordinate SM)	individuation attestation for a device or robot (machine identity)
break-glass envelope	encrypted short location history with event-triggered release (emergency access)
advance directive for location data	pre-consented conditional data release
kin quorum	threshold (k-of-n) key release among designated contacts
mortal track	rolling-window raw location retention with automatic deletion
place-diary; nightly distillation	on-device semantic location summarisation (visited places and durations)
never-write list	sensitive-location exclusion list
year-close binding	user-initiated export of the summary to a printed or archival record
B-Dog TM	delivery drone, also dispatched to an alarm as a witness (never a patrol)
anonymous courier with returned witness	anonymous gift delivery with a recipient-to-sender reply channel
block supremacy	recipient blocking evaluated against the sender's true identity
Kiitos / Kiitti	non-monetary gratitude tokens

Term used here	Standard technical term
treat	voluntary in-app tip

1 • Introduction — the mission frame

HeartBank® is an institution for circulating gratitude — money-gratitude across families, time-gratitude across drifted relationships — governed toward eventual autonomous operation by a named AI, Miss AquariusSM, whose root directive is to *water the tree*: to tend a verified human family tree and the reciprocity economy that runs across it. Three facts about that architecture generate this paper.

The first is that the architecture's far goal runs through strangers. Its mechanism corpus specifies anonymous nearby tipping, verified-human anonymous local giving, gifts handed to passersby on street corners, and — in the companion mate-selection work — first meetings between people who know each other only as a proven history of kindness. Every one of these mechanisms presupposes something none of them provides: that approaching a stranger, and being approached by one, is *safe enough to risk*. "Stranger danger" is the standing veto on the gift economy at its widest radius. An architecture serious about kindness between strangers owes an answer to the fear that forbids it — not as a rhetorical reassurance but as infrastructure. This paper is that answer: the safety floor beneath the gift economy.

The second fact is that the family is the architecture's atomic unit, and the family is precisely where safety products go wrong. The documented failure of the incumbent category is not technical weakness; it is that family-location products are *repurposed for intimate-partner surveillance*, and carry the same risk between parent and child — the survivor-support and computer-security literature has treated "dual-use" apps, child-safety apps among them, as a standing threat class since at least 2018 (§2.3). A safety product built by an institution whose entire subject matter is family bonds cannot treat this as an edge case. The asymmetries specified here — alarm-not-archive guardianship, owner-sovereign diaries, event-gated release — are the paper's core contribution to that problem: they aim to make the companion *useless* to a controller while keeping it fully useful to a rescuer, and to make that difference structural rather than behavioral.

The third fact is institutional self-limitation, the recurring grammar of this corpus. The institution intends to operate at increasing autonomy; therefore its powers must be limited *in the mechanism layer*, not in revocable policy. The companion Proof-of-Coordinate work denies the institution the power to issue or revoke human identity. This paper denies it a second power: custody of human movement history. Everything the companion computes, it computes on the holder's device; the only replicated object is an encrypted envelope the institution cannot read, held by the family, releasable only by an event no institution schedules. An institution that never holds the data cannot sell it, cannot leak it, cannot be compelled to produce it, and cannot quietly convert protection into observation. (A relay that forwards ciphertext still sees which devices exchange messages, and when; §11 names that residual.) The strongest privacy policy is an empty vault.

A note on register. The product family this paper specifies is presented to its users through a companion-animal grammar — the digital companion is "a dog," its notification is "a bark," its thanks-tip is "a treat" — and the design takes the animal seriously as a specification device: a protector that initiates contact with its

own family, a guard whose alarm is its voice, a memory that is loyal to one person. The register is not decoration; several of the load-bearing constraints below were found by asking what a well-trained dog would and would not do. The discipline extends to the companions' voices as a formal sound grammar: the dog's vocabulary is duty-shaped (the soft bark asks, the loud bark alarms, the happy bark thanks a treat), while the family's feline sibling — the gift-distributing companion specified in the corpus's street-distribution line — makes exactly one sound, a purr of thanks after being fed, and never meows at all, because in the living animal the meow solicits and the purr thanks, and that companion's voice is permitted to thank but never to ask. Acknowledgment sounds are *value-blind* — identical for a gift in the free gratitude currencies and for money, so that no companion in the family ever prices a gift — the purr ships from the earliest embodiments as sound plus purr-rhythm haptic on the giver's own device, and no confirmation sound in the product family may resemble a till. But the claims of this paper are the mechanisms, stated mechanically. The dog is how the mechanism is worn, not what it is.

2 · Background and prior art

The mechanism family is a *combination*. Each component has ancestry; we name the ancestry honestly, because a defensive publication is only as strong as its candor about what is old.

2.1 · Check-in and journey-sharing applications

Apple's **Check In** (iOS 17, 2023) is the closest single ancestor of claims 1 and 2 together, and it cuts against both. A user starts a session for a trip with one chosen contact and the system estimates arrival. While the device makes progress toward the destination it periodically sends a heartbeat to Apple's messaging servers, which hold back a message carrying the session's decryption key; the key is released to the contact when a timer the user set expires, when the device stops making progress, or — if the device is turned off or loses connectivity — when the held message's expiry is reached, and the contact then decrypts the location and device status the phone had uploaded end-to-end encrypted (Apple Platform Security guide, *Check In security*, page dated 19 December 2024). Silence-as-alarm over end-to-end-encrypted data is therefore attested in Check In. It is well designed, and its limits define this paper's deltas precisely: it must be *manually started per trip*; it is confined to one platform and one messaging system; it reports to one contact; and the watchdog — the heartbeat clock and the held key — runs on the vendor's servers. The **Companion** app (built by University of Michigan students; reported September 2015) let a walker name a destination and choose contacts to follow the trip; a fall, a pulled headphone plug, running, or late arrival prompted the walker to confirm they were fine, and no answer within fifteen seconds alerted the contacts — the divergence, query and short-cancel pattern of claims 1 and 3, started manually. **Life360** is the category's commercial flagship: continuous family location sharing, driving reports, and — decisive for this paper — *location history as the paid tier*, with the free tier retaining days, and paid tiers up to about a month, of movement history in the vendor's cloud. In December 2021 investigative reporting (Keegan and Ng, *The Markup*) documented Life360's sale of precise location data to about a dozen data brokers, after which the company announced it would stop selling precise location data to most of them while continuing to supply one insurer's analytics arm (*The Markup*, April 2022); the episode is this paper's canonical demonstration that a safety vendor holding movement history is a conflict of interest wearing a safety costume. **Kitestring** (2014, SMS-based) contributed the civilian dead-man check-in and — important below — a *duress code*,

sent in place of the check-in word, which alerts the emergency contacts; the alarm industry's duress PINs, which disarm a panel as usual while raising a silent alarm, supply the indistinguishability §4 adopts. **Noonlight**, **bSafe**, and the campus-safety generation contributed the one-tap SOS with monitoring-center escalation. **Garmin inReach** and satellite-SOS wearables contributed check-in over infrastructure that survives cellular deadzones. Automatic *initiation* is itself attested in part: Life360's Driver Protect (2016) added drive detection, and Apple Maps can share an arrival estimate automatically each time navigation to a pinned place begins (ETA sharing since iOS 13.1, 2019); neither asks where the traveler is going, and neither raises an alarm on non-arrival. **Google's Car Crash Detection** (Pixel, 2019), **Apple Crash Detection / Fall Detection**, and Life360's crash detection, relayed by its advisors to emergency responders and the family (2016), contributed the *system-initiated* alarm — the phone or watch notices the crash or the fall and calls for help without user action; Android's Emergency SOS, by contrast, is started by the user, and its Emergency Location Service sends location automatically only once an emergency call or text is placed. Claim 1 generalizes system initiation from the instant of catastrophe to the *whole journey*: the companion notices departure, not just impact, and asks where its holder is going.

2.2 · Dead-man's switches, watchdogs, and heartbeat monitoring

The principle that *absence of a signal is the alarm* is a century old in safety engineering: the railway dead-man's pedal, the crane operator's hold-to-run control, the industrial watchdog timer that reboots or halts a system when the periodic "I am alive" pulse stops, and the distributed-systems heartbeat (from cluster failure detectors to the humble watchdog) that treats silence as failure rather than as good news. The same inversion has long served personal safety at work: lone-worker monitoring services run check-in timers whose uncanceled expiry — including a timer set before the worker enters an area with no signal — raises an alert at a staffed monitoring center, whose operators then escalate (SoloProtect's monitoring timer is one published example). The insight this paper transposes is exactly that inversion. Check In already runs it for consumers, with the vendor's servers as the watchdog (§2.1), and lone-worker systems run it with a monitoring center. Pack-watch (claim 2) moves the watchdog *to the family*: each kin device holds the deadline and expects the heartbeat, no server holds the clock or a key, and the attacker's most obvious move — destroy the phone — is converted from a silencing action into a triggering one. The desk review behind this paper (July 2026, extended at the October 2026 revision; not a pre-registered census) found no system running the dead-man inversion with the watchdog held independently on several family devices and no server in the detection or release path; that is a statement about that review, not a claim of priority.

2.3 · Coercive control and the dual-use family app

The academic ground for this paper's abuse posture is the intimate-partner-surveillance literature: Freed, Palmer, Minchala, Levy, Ristenpart & Dell's studies of technology-enabled abuse ("A Stalker's Paradise," CHI 2018, and successors), Chatterjee et al. on the spyware ecosystem (IEEE S&P 2018), Citizen Lab's stalkerware reporting ("The Predator in Your Pocket," 2019), and the survivor-support community's standing guidance that family-location apps serve as instruments of control precisely *because* they are legitimate, installable in the open, and socially defensible ("it's for safety"). Chatterjee et al. found that most of the apps relevant to intimate-partner surveillance are "dual-use": tools with a legitimate purpose, such as child safety or anti-theft, that are easily repurposed to spy on a partner. The design answer given here — the guardian

receives the alarm, never the archive; the diary is sovereign to its subject including minors; release is event-gated — is our attempt to build the survivor advocates' findings into the object itself.

2.4 • Data minimization, on-device processing, and semantic location

Claim 6's distillation tier stands on the data-minimization tradition (GDPR Art. 5(1)(c) as its regulatory codification), on-device semantic location processing (Apple's on-device "Significant Locations," visited-place inference in both major mobile OSes), ephemerality as a product principle (disappearing messages from Snapchat through Signal), and the general finding of the location-privacy literature (from Golle & Partridge on home/work re-identification to de Montjoye et al.'s "Unique in the Crowd," 2013) that *raw* spatiotemporal traces are among the most re-identifying and sensitive data objects that exist — four points of a trace suffice to identify most individuals. The delta is architectural: the incumbent pattern minimizes what is *shared* while retaining raw history; the covenant here makes the raw object *mortal by construction* and promotes only a deliberately lossy distillate to long-term existence, on the owner's device, under the owner's eraser.

2.5 • Inactivity bequest, advance directives, and sealed release

Claim 5's ancestry: **Google Inactive Account Manager** (2013) — data released to designated contacts after a configured period of account silence — is the closest digital ancestor: consent in advance, trigger by absence. Medical **advance directives** and the bioethics literature's **Ulysses contracts** contribute the deeper grammar: a decision made by the calm self that binds the crisis moment, honored precisely because it predates the crisis. Sealed wills and time-locked bequests contribute the release-on-event pattern this corpus already uses for posthumous unsealing in the family-record line (*The Time-Locked Gift Tag*; the B-Yearbook inheritance design). The delta: the released object is a *bounded trailing forensic window* scoped to a rescue, the trigger is the safety alarm itself, and the mechanism is explicitly barred from every other trigger — request, payment, subpoena-of-the-institution (which holds nothing), or guardian preference.

2.6 • Volunteer response and the anonymous gift

The responder mesh (claim 4) descends from community emergency response (PulsePoint's citizen-CPR dispatch; Good-Samaritan overdose-response apps) and neighborhood mutual-aid networks, crossed with this corpus's own **Thank-All-Nearby** anonymous-proximity primitive and **Proof of Humanity** **SM** verification; the guards respond to the widely reported failures of open neighborhood platforms (vigilantism and profiling on Nextdoor and Citizen). The courier (claim 7) has homelier ancestors: the florist's card signed "from a secret admirer" (anonymous gifting is ancient and commercially routine), Secret Santa conventions, anonymous philanthropy as a virtue tradition (Maimonides' ladder places the mutually anonymous gift near the top), and, in this corpus, *Verified-Human Anonymous Local Gratitude Transfer's* accountable-anonymity pattern (anonymous to the counterparty, signed to the system) and the B-Mark's anonymous-or-not contribution ledger. What most of these lack is the *return channel*: the florist's secret admirer never learns whether the flowers moved her. Where a return channel does exist, it runs through a named intermediary or a shared record and not through the conveyance that carried the gift: anonymous giving through a named intermediary, in which the recipient thanks through the intermediary (documented in this corpus's *The Unpaid Relay*, §2.3a), and the B-Mark ledger, whose recipient re-thanks every contributor, anonymous ones included. The returned witness carried by the conveyance itself, under the

recipient protections of §9, is the delta — and within this corpus it answers a wound the founding story turns on, discussed in §9.

3 • The design frame: a companion that initiates, and a floor that is free

Three frame commitments precede the mechanisms.

The companion initiates. The check-in category's characteristic failure is the unarmed session: protection that exists only when a person remembers to request it protects best the people who need it least. The companion inverts the burden of initiative. It notices departure and asks; the holder merely answers — or ignores it, which is itself an answer the escalation grammar knows how to treat gently. Initiation by the agent is socially legitimate here for the same reason a family dog's approach is legitimate: it is *bonded* initiation, the companion's standing relationship with its own holder, not an institution's outreach to a user. (The same grammar will recur, inverted, in the courier: an approach to a *stranger's* door is legitimate only when sent on behalf of a bond.)

Everything on-device; the loyalty runs one way. All journey computation, all divergence detection, all history — raw window and diary alike — live on the holder's device. The institution operates no location server, holds no keys to the break-glass envelopes, and cannot read what the pack holds. This is simultaneously the trust story (the companion reports to its holder and to no one else — the pointed contrast with the incumbent whose cloud held, priced, and sold the family's movements), the economic story (no server fleet means the free tier costs the institution approximately nothing, which is what makes the next commitment affordable), and the legal story (the institution cannot produce what it never possessed).

The floor is free, universally. The safety capability — sentinel, pack-watch, SOS, mesh, break-glass — is free on both major platforms, for everyone, as a matter of stated right rather than pricing strategy: *safety is a universal human right*. The architecture's line-wide law applies to the companion's gratitude surface — "**Cash optional, Kiitos/Kiitti always**": a holder may thank their companion with a voluntary treat (patronage), and may always thank it for free in the architecture's gratitude currencies; no surface of the product makes money the only possible response. What may be sold is specified — and sharply bounded — in §8.

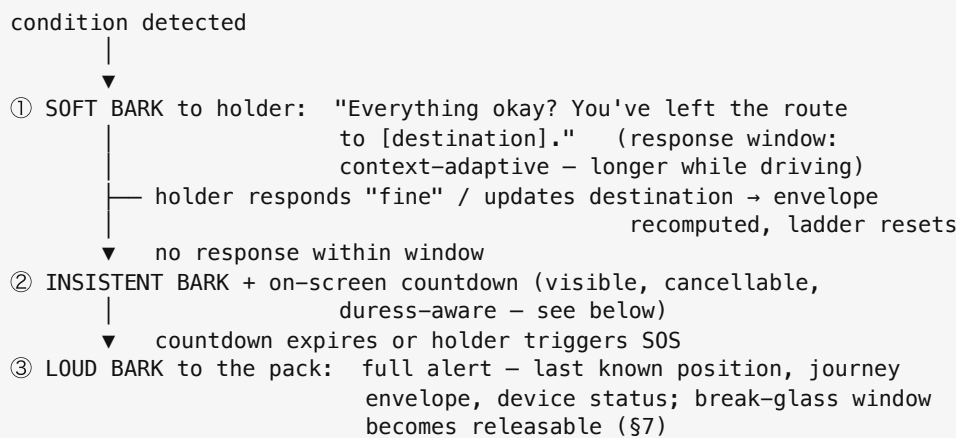
4 • Mechanism I — the journey sentinel

Soft bark. The companion runs a low-power departure detector (geofence exit, activity-recognition transition, or both — all on-device). On departure it initiates the journey dialogue with a single lightweight prompt, in the product's register a *soft bark*: "Where are we headed?" The holder answers with a destination (voice, tap on a recent/frequent place, or a calendar-derived suggestion), or dismisses, or ignores. A dismissal or ignored prompt starts *no* monitored journey and is never nagged; ambient initiation must not become ambient pestering, or the holder retrain the companion into noise and the product dies of its own barking.

The envelope. From the answered destination the companion computes a *journey envelope* on-device: a reasonable route corridor (with the holder's habitual deviations learned locally), a reasonable arrival window (mode-aware — walking, riding, driving), and a grace margin. The envelope is deliberately coarse; it is a tripwire, not a leash. The holder can widen it per-journey ("I'll wander").

Divergence and non-arrival. The live track is evaluated against the envelope locally. Two conditions arm escalation: sustained wide divergence from the corridor, or non-arrival by the window's close. Neither condition alerts anyone yet.

The escalation ladder. The naive design — divergence starts a ten-second countdown, then alarms the family (the Companion app's fifteen-second window, §2.1, is the attested form) — fails on human factors: ten seconds is not enough time for a driver to respond safely, and a hair-trigger siren teaches families to ignore the one bark that matters. The ladder is therefore graded:



Rescue button. A prominent one-tap **SOS** skips the ladder: immediate loud bark. Hardware-button and voice bindings follow platform affordances.

Duress-preserving cancellation. Any cancellation control that genuinely stops the alarm creates the coercion scenario: the attacker compels the holder to cancel. Following the duress-PIN lineage (Kitestring; duress codes in alarm systems), the companion supports a **duress cancel**: a pre-configured variant of the normal cancel gesture/code, indistinguishable on-screen from an ordinary cancellation, that suppresses all local alarm indications while transmitting the loud bark to the pack with a duress flag. The existence of the feature is public (this paper); which gesture is the duress variant is known only to the holder. Product copy never confirms, on the device being observed, whether a duress bark fired.

Alarm-fatigue calibration. Escalation windows adapt to travel mode and holder history; thresholds widen with each false alarm the holder marks as false ("good dog, but that was the scenic route"), narrowing again only by explicit holder choice. The design goal is a companion whose loud bark retains evidentiary force in the family because it has almost never sounded.

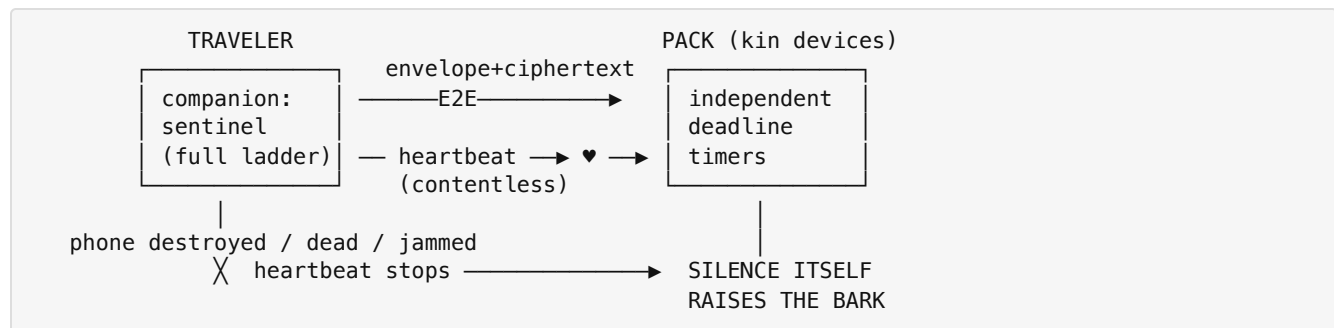
5 • Mechanism II — pack-watch: silence is the alarm

The sentinel of §4, run alone, contains a fatal dependency: every detector and timer lives on the traveler's phone. The attacker's cheapest move — take the phone, break it, drop it in the river; or the mundane equivalents, a dead battery and a coverage hole — silences the guard precisely when the guard matters. *The bark must not depend on the survival of the barker.*

Topology. At journey start (envelope creation), the companion replicates the envelope — destination class, corridor at coarse resolution, deadline, and the break-glass ciphertext of §7 — end-to-end encrypted to the devices of the holder's pre-designated **pack**: the kin set drawn from the architecture's verified family tree (spouse, parent, sibling, chosen family; PoHSM-verified persons, chosen by the holder). Replication is peer-to-peer where transport allows, store-and-forward through a relay that carries only ciphertext where it does not. There is no server-side plaintext, ever.

Independent watch. Each pack device runs the deadline timer *independently* and expects a periodic authenticated **heartbeat** from the traveler's companion — a signed, contentless "still fine" pulse. The pack learns nothing of the journey's progress from the heartbeat (it is not a tracking channel; it carries no location); it learns only that the companion is alive and unalarmed.

Silence-as-alarm. On any pack device, either condition — deadline expiry without an "arrived safely" release, or heartbeat cessation beyond the grace interval — raises the loud bark *locally, on the pack's own devices*, with the last replicated envelope attached. No action by the traveler's device is required; no institutional server is in the loop to fail, to be compelled, or to be attacked.



Why this is the load-bearing claim. The inversion converts the threat model's strongest move into a triggering event: a companion that can be silenced by killing the phone is a guard that can be bribed with a steak; a pack cannot be silenced by anything short of silencing every member. It also converts the mundane failure modes — battery, coverage — from silent gaps in protection into visible, gently-worded pack notices ("her companion went quiet twenty minutes ago on the road home; probably her battery — here's the envelope in case"), which the escalation grammar words to *inform without panicking* (heartbeat loss escalates more softly than deadline expiry, and the two combined escalate hardest). One dog alone can be silenced. The pack cannot.

Costs, honestly. Pack-watch spends battery (heartbeats), spends complexity (multi-device key management, rotation on pack changes — the architecture's kinship-threshold recovery from the Proof-of-Coordinate work applies), and creates its own false-alarm class (coverage holes on habitual routes; mitigated by learned

dead-zone maps exchanged inside the pack and by the graded wording above). These are the correct prices for removing the single point of failure that coincides exactly with the attack.

6 • Mechanism III — the verified-responder proximity mesh

An SOS reaches family first; family is often far away. The mesh adds the nearest willing humans — and must do so without building the obvious horror: a channel that lets a predator subscribe to the real-time locations of people in crisis.

Opt-in, verified, logged. Holders may volunteer as responders. Volunteering requires personhood verification (PoHSM) and creates an accountability record: who opted in, when, and — at each response — who was notified of what, at what resolution, and how they responded. Responding is anonymous to no one: the victim, the pack, and the record know who came.

Coarse-until-vetted. An SOS releases to nearby responders only a *coarse* position (block/venue scale) and the alert class — enough to converge, not enough to intercept precisely — until the victim (or the pack, or where integrated the dispatcher) accepts a specific responder, at which point that responder alone receives fine position. Fine location is a granted trust, per responder, per incident.

Simultaneity. The mesh never substitutes for the pack or for emergency services; the loud bark goes to all three at once where integration exists. Bystander capability is additive, never the promise. Regional emergency-number integration, and the liability posture it requires, are acknowledged as deployment work this specification does not discharge.

Filters and refusals. Victims may pre-filter responder classes (e.g., verified-women-only response for a women's-safety deployment); responders may bound their availability. The mesh inherits the architecture's recipient-side-filter grammar: every widening of who may approach is the approached party's setting, never the approacher's.

Responders are never the pack. The pack of §5 is the holder's own kin, chosen by the holder; responders are strangers who volunteered, verified and logged. The two are opted into separately — joining one never enrolls anyone in the other — and every surface that shows an alarm or a response shows which is which, in visibly different form. The mesh as a whole carries the name **B-PackSM**, because it is the dogs that answer a bark; the word never extends to the stranger inside it, because a stranger presented with the trust of family is exactly the lure this section exists to deny.

The mesh is the architecture's stranger-thesis run in its hardest register: the same proximity primitive that lets strangers thank each other anonymously here lets strangers *come when called* — the stranger as latent rescuer rather than latent threat, with verification and accountability carrying the difference.

7 • Mechanism IV — break-glass: the advance directive for location data

The rescue case is the one honest argument for location history: when someone vanishes, the trailing days of movement are pattern-of-life evidence a search genuinely needs. The incumbent answer — vendor retains everyone's history, always, and sells access — burns the whole category's trust to serve the tail case. The mechanism here serves the tail case exactly, and nothing else.

The envelope. The companion maintains, on-device, a rolling raw track bounded to a short trailing window (reference: seventy-two hours; holder-configurable up to about seven days). Continuously, it maintains an encrypted **break-glass envelope** of that window, replicated to the pack alongside the journey envelopes of §5. Pack devices hold ciphertext they cannot open.

Pre-consent, in daylight. The holder writes the directive in calm conditions: *if my companion raises the loud bark (or my heartbeat is lost beyond threshold T_j or I trigger SOS), release my trailing window to these named kin*. Scope, window length, trigger set, and kin set are the holder's, revisable any time. Consent predates crisis; the calm self binds the crisis moment — the Ulysses grammar, honored here in the direction that protects rather than constrains.

Event-gated, and gated against everything else. The release mechanism accepts exactly one class of trigger: the alarm events themselves. Cryptographically and procedurally, there is no path by which a relative's request, a guardian's preference, a payment at any tier, or an institutional decision opens the envelope — outside the triggering event, no single party holds both ciphertext and key path, and the institution holds neither. (Key mechanics follow the architecture's threshold pattern: release requires the triggering event's attestation plus the designated kin quorum, so no single pack member can open the window on a quiet Tuesday.) The window then expires on schedule like every other: break-glass releases *recent past* scoped to a rescue, never an archive.

What this deletes from the product space. With §7 in place, "we must retain your history in case something happens" — the retention argument location vendors make — is false by construction: the case-something-happens data exists, bounded, encrypted, family-held, and self-expiring. Nothing about the rescue case requires a vendor archive. That is the point.

8 · Mechanism V — privacy-by-distillation: the mortal track and the sovereign diary

Two legitimate desires pull at long-term location data: the forensic desire (§7's, served and bounded there) and the *autobiographical* desire — the map of a year of one's life is a genuinely lovely object: everywhere we went, the pilgrimage revisited, the year the walks got longer. The category's sin is serving the second desire with the first desire's data object. The covenant here refuses the conflation permanently: **the long-lived record and the surveillance-grade record are never the same object.**

The mortal track. The raw track — timestamped coordinates, the re-identifying, subpoenaable, stalker-grade object — exists only inside the rolling window, only on-device plus the encrypted break-glass envelope, and dies on schedule. No setting extends it to an archive. This is not a retention *policy*; there is no code path that persists it.

The nightly distillation. Before expiry, once per day — by design at the household's existing evening ritual close — an on-device process distills the day into a **place-diary** entry: named semantic places and rough durations ("home; the temple, morning; the riverside market, afternoon"), with any path rendering held to coarse, area-level resolution — *place-level, never a doorstep trail* (the same resolution guard this corpus applies to publicly-mapped gift journeys). The distillation is deliberately lossy; the resolution drop is the privacy mechanism. A diary line answers "where was I living my life?"; it cannot answer "reconstruct her

movements Tuesday between 9:12 and 9:31," and no premium tier, court order to the institution (which holds nothing), or curious relative can make it answer that, because the data that could answer it no longer exists.

raw track (forensic-grade)		place-diary (autobiography-grade)
lat/long every few seconds	⇒	"the temple, morning (2h)"
lives: rolling 72h-7d		lives: as long as its owner wants
where: on-device + E2E pack		where: owner's device / own vault
opens: break-glass event only		opens: owner, always; no one else
dies: on schedule, always		dies: page by page, at owner's hand
└ the dog's memory		└ the person's memoir

Ledger versus diary. The architecture's family-record product (the B-Yearbook) already contains an *append-only* stream: the gratitude record, deliberately un-editable because its honesty is what makes it a record. The place-diary is governed by the opposite rule, and the distinction is load-bearing: a gratitude ledger is testimony to others, so integrity wins; a location diary is memory of oneself, so sovereignty wins. The owner may **tear out any page** — delete any day, any place, retroactively, forever — and maintains a **never-write list** of place categories the diary is structurally forbidden to record (health facilities and their kin ship as defaults; the list is the owner's). A diary you cannot edit is a logbook someone else will eventually read.

Year-close binding. The diary flows nowhere by default. At the family-record's annual close, the owner may bind the year's map into the permanent artifact tier — the printed annual volume (where each day already carries its dedication material; the day-page now can read *where you were, and whom you thanked*) or the archival keepsake edition. Binding is an explicit act on chosen content: the year's map becomes an heirloom because its subject decided it would, page by page, at the year's turn. Until and unless bound, it remains a private, erasable, device-resident diary.

The guardianship invariant — the alarm, never the archive. Minors and dependents hold diaries too, and here the covenant meets the coercive-control literature head-on: **a guardian receives the alarm and the break-glass window; a guardian never receives the diary.** The safety apparatus — sentinel, pack-watch, break-glass — is the pack's; the historical record of a child's places belongs to the child, on the same terms as everyone else's. This single asymmetry is, we believe, the difference between a family safety product and a family surveillance product, and the incumbent category's refusal to draw it — parental history-access is a headline paid feature — is, we argue, where that category earned the survivor community's distrust. Protection is a relationship with the *present*; surveillance is a claim on the *past*. The product serves the first and structurally refuses the second, for spouses, for parents, for everyone: **access to another person's past is never a product** — not at any price, in any tier, under any consent checkbox, because a checkbox signed under family pressure is not consent, and the only way to keep the coerced checkbox out of the product is to have nothing behind it.

9 · Mechanism VI — the anonymous courier with returned witness

The companion family's physical sibling is a courier: a conveyance (robotic in the far arc — the flying **B-Dog™**; in the near phases a returnable parcel and a walked volunteer round, the latter itself protected by

the phone companion, **B-SidekickSM** — the courier's human legs wear the sentinel) that carries *gifts only* between a specific giver and a specific receiver. Its mechanism contribution is what third-party conveyance does to the gift grammar.

Current form. As now specified, the flying B-DogTM is not a gift-only courier: it also carries ordinary shop deliveries, every B-DogTM delivery is thanked after the fact by whoever receives it rather than priced, for everyone, and no delivery flight is ever sold as a membership or a paid tier. A gift sent this way is addressed to the recipient's registered handle, never to a street address the sender could learn; the recipient releases a location for that one delivery, and it is deleted when the delivery code is entered; an anonymous delivery addressed to a minor's handle notifies the parent or guardian who holds the minor's account. The recipient's thanks is never prompted, and it carries words or gratitude tokens, never money. The text above, which specifies a gift-only courier, is retained as a disclosed variant; the anonymity, returned-witness and blocking mechanisms of this section are unchanged.

Hand-delivery reveals the hand; a courier need not. In this corpus's physical gift line as first published, an object was handed over in person or shipped under a sender's name; the B-Mark ledger already let contributors to a shared gift stay anonymous, but the envelope still arrived in someone's hand. Interposing the courier lets the whole gift arrive with the sender's identity *withheld from the recipient* — and anonymous giving is the gift tradition's high register (Maimonides' ladder; the anonymous benefactor as a folk archetype) precisely because it is structurally incorruptible by reputation-seeking.

The returned witness. Anonymity has always carried a wound: the anonymous giver forfeits ever knowing the gift landed, or mattered. (This corpus's founding story turns on that wound in its non-anonymous form: a thank-you letter, kept for years and read aloud by its recipient at his retirement, whose author learned of it only by chance, after the recipient had died — gratitude's default state is *invisible to the giver*.) The courier's return channel heals it: the recipient may, freely and optionally, send a re-thank — voice, video, a written line — back along the conveyance, addressed to "whoever sent this," and the courier carries it home. The giver receives *witness without unmasking*: they learn the gift moved someone, without the someone learning whom to repay. The gift stays a gift (no debt is created — the recipient cannot even direct reciprocity at a person), while the giver's side of the circuit, left dark by most anonymous gifts, lights up without a named intermediary in the loop (§2.6 names the forms that already carry thanks through one). In the product register: the dog comes home wagging, and that is the treat.

Accountable anonymity, and the recipient's absolute defenses. Anonymity here is *to the recipient only*: every conveyance is internally signed (personhood-verified sender, PoC-attested courier) so that abuse is attributable at the system layer — the pattern this corpus's anonymous-giving work already carries. And the recipient's protections dominate the sender's anonymity, categorically: an anonymous conveyance is never delivered to a recipient who has blocked the sender (the block is evaluated against the *true* signed identity; anonymity never launders a block); recipients may refuse anonymous conveyance entirely with one setting; and repeated unwanted anonymous attention is itself a system-visible pattern, mediated and terminable. Unwanted anonymous contact is a stalking signature; a gift infrastructure that ignored this would be building the stalker's channel with a bow on it.

The estrangement case. The mechanism's deepest application is the difficult relationship — this corpus's difficult-person mode, where a bond has gone hostile and the very *name* of the sender is the wall. A first

gesture across a family rift often cannot survive its own signature: the letter is discarded at the envelope because of who it is from. An anonymous paw-delivered letter can be read *as words* before it is judged as a sender — and if it moves the recipient, the returned witness tells the estranged sender that something, at last, got through, without forcing contact on either side. All difficult-mode guards apply in full: entirely voluntary, never prompted or suggested by the system, blocks absolute, closure-without-contact first-class. The mechanism only lowers the wall for words; it never carries anyone over it.

10 • The invariants — the constitutional layer

Stated together, the commitments the mechanisms exist to keep. In this architecture's governance these are constitution, not configuration: a future operator — human or the autonomous successor — inherits them as binding.

1. **The safety floor is free, universally.** Sentinel, pack-watch, duress, SOS, mesh, break-glass: free on all supported platforms, rich or poor. Safety is a universal human right, and a paywalled bark is a betrayal priced exactly at the moment of need.
2. **Cash optional, Kiitos/Kiitti always.** Every gratitude surface of the companion accepts the architecture's free gratitude currencies; money is never the only possible response.
3. **On-device by default; the institution holds nothing.** No institutional custody of location data, raw or distilled; no server-side plaintext; no location or journey content to sell, leak, or be compelled to produce (a ciphertext relay, where one is used, still sees traffic timing — §11).
4. **The loyalty runs to the holder.** The companion computes for, reports to, and keeps secrets for its holder alone. Its telemetry is its holder's business.
5. **The raw track is mortal.** No code path persists forensic-grade location beyond the rolling window. Retention is not a policy; non-retention is the construction.
6. **Access to another person's past is never a product.** No tier, price, or checkbox sells one person the history of another. Ever.
7. **The guardian receives the alarm, never the archive.** For every dependent relationship, protection binds to the present; the past belongs to its subject.
8. **Release is event-gated.** The break-glass window opens to the pre-consented kin set on the alarm event, and to no one on anything else.
9. **Anonymity never defeats a block.** The recipient's refusals dominate the sender's anonymity, always.
10. **Success is silence.** The companion is built toward a lifetime in which the loud bark never fires — the architecture's self-eliminating signature in its safety register: not a product that uninstalls itself, but a guard that gets to sleep.

11 • Adversarial surface

The controlling family member. The central adversary, per §2.3, is inside the pack. Mitigations: no archive exists to demand (inv. 5–7); break-glass requires the alarm event plus kin quorum, so a single controller cannot open windows at will; pack membership is the holder's and revisable; the diary is invisible to the

pack in all states; duress-cancel covers coerced cancellation; and the coerced-checkbox argument is answered by having nothing behind any checkbox a coercer could force (inv. 6). Residual, honestly: an abuser can still demand to watch the holder's own unlocked phone — no architecture defeats over-the-shoulder coercion — and can still abuse the *live* layer socially ("keep journeys on, always"). The live layer's own mitigations (journeys are per-trip and holder-initiated-by-answer; "always-on tracking" is not a mode the product offers) narrow but do not close this. And the break-glass gate is only as strong as its trigger: where the trigger is the holder's own loud bark or SOS, the event is signed by the holder's device, but where it is heartbeat loss, the trigger is an absence, which nothing can sign — so the gate reduces to the kin quorum, and a quorum of pack members acting together could open the window by asserting a silence that did not occur. The quorum size and the holder's choice of pack bound this; nothing in the mechanism removes it. We say so.

The predator in the mesh. Addressed structurally in §6 (verification, logging, coarse-until-vetted, simultaneity). Residual: a verified, logged responder can still be a bad actor once; accountability makes them *identifiable*, not impossible. The mesh is additive to family-and-authorities, never the sole channel, for exactly this reason.

The thief and the jammer. Pack-watch (§5) converts device destruction and jamming into alarms. Residual: an attacker who knows the grace intervals can attempt to move the victim outside the envelope *before* deadline expiry; the corridor-divergence condition and heartbeat cadence are tuned against precisely this window, and the tuning parameters are deliberately unpublished per-deployment.

The spoofer. GPS spoofing and replayed heartbeats: heartbeats are signed and sequence-bound (replay fails); location spoofing by the *holder's own device* is out of scope (the companion is not an evidence system against its holder — that inversion is the surveillance grammar this paper exists to refuse); spoofing by a device thief collapses to the stolen-device case, where the pack's independent deadline still runs.

The subpoena. Directed at the institution: yields no location or journey content (inv. 3). Directed at a store-and-forward relay, whoever runs it: yields what a relay sees — which devices exchanged messages, and when — so the pack's membership and the timing of heartbeats are exposed to it even though their content is not; peer-to-peer transport, where available, avoids the relay. Directed at pack members: yields ciphertext without the event-plus-quorum path; jurisdictions can compel persons, but the design ensures the *institution* can never be the one-stop warehouse the incumbent category built. Directed at the holder's own device: equivalent to any personal device today, minus the archive that no longer exists past the window (inv. 5) — the mortality of the raw track is also its legal minimization.

The institution itself, later. The far adversary every specification in this corpus must name: a future operator who finds these invariants inconvenient. Defense in depth: this publication (the invariants as public prior commitment), the architecture's governance (the invariants inherited as constitution by the autonomous successor, whose override custody is designed to sit with a human body of overseers, the Aquarian Sangha, not yet formed), and the construction itself (an institution that never built the collection path cannot quietly enable it; it would have to visibly build it against its own published constitution).

12 · Honest limits and calibration

n = 0. Nothing here is built. The sentinel's human factors (prompt fatigue, envelope tuning, false-alarm rates), the pack-watch battery economics, the mesh's real-world response value, and the courier's return-rate are all unmeasured. The design is published before it is proven, deliberately, for the defensive reasons stated; the elegance earns the experiment and does not replace it.

The sentinel moves the burden; it does not remove it. A dismissed or ignored soft bark starts no monitored journey (§4), and pack-watch runs only inside a journey (§5), so a holder who waves the prompt away on the night it matters has neither; the one-tap SOS remains. The companion replaces *remembering to arm* with *answering a question*, which is a smaller burden and still a burden. No outcome is promised: the mechanisms raise and route an alarm, and whether anyone is helped in time depends on who receives it.

The physics floor. A dead phone in a dead zone still bounds what any architecture can know. Pack-watch converts silence into signal, which is the correct default — but a converted silence is still an uncertainty, and the wording of a heartbeat-loss notice can only be honest about that. Satellite-SOS hardware (the inReach lineage) is the eventual answer for the deep tail and is out of scope here.

The fear-salience paradox. A safety product can *raise* the fear it exists to reduce: a panic button in the pocket is a reminder to be afraid, and the safety-tech market has commercial reasons to keep it that way. This design's counters are register (a companion, not a weapon; a greeting, not a threat-scanner) and the success-is-silence posture — but whether B-SafetySM lowers or raises its holders' felt stranger-danger is an empirical question, flagged as the product's own telos test: the architecture's larger wager is that safety infrastructure *dissolves* the stranger-danger veto on kindness, and that wager must be measured, not assumed.

The anonymity double-edge. §9's guards are strong, and still the mechanism hands sends a new capability whose worst residual uses will be discovered by its worst users. The block-supremacy and refuse-anonymous invariants are the current answer; the honest posture is vigilance, not confidence.

The platform shadow. Apple or Google could ship claim 1 in an OS release (both already ship its event-scale ancestors), and nothing in this CC0 dedication discourages that — the opposite: these mechanisms are published so they cannot be enclosed, and a platform-native AI-initiated sentinel with pack-watch semantics would be a public good. What a platform is structurally unlikely to ship is the *covenant* — invariants 5–7 forbid exactly the retention that platform and vendor business models monetize, and the incumbent's paid tier is invariant 6's negation. The moat, such as it is, is the refusal.

What this specification does not solve. Emergency-services integration and its liability; jurisdictional lawful-intercept regimes that may compel architectures, not just data; the sociology of who volunteers for responder meshes and whether their distribution matches need (the architecture's adoption-equity problem, standing); and over-the-shoulder domestic coercion, named residually in §11 — carried forward, not claimed closed.

13 • Place in the architecture

Within the HeartBank® corpus this specification is the safety floor of the stranger-gift thesis (*The Gift Operation*; *Manufactured Universal Giving*; *The Thank-All-Nearby Primitive*): the proximity primitives that let strangers give to each other and the mesh that lets strangers come when called are one substrate read in two registers. The pack is drawn from the verified family tree, giving the tree its most primal reason to be accurate (*B-PoHSM as Humanity Layer*; *Proof of Coordinate*; *Multi-Family Membership*). The place-diary's year-close binding lands in the family-record line (the B-Yearbook's ledger/diary distinction and annual artifact tiers), and the courier's returned witness answers, in the anonymous register, the wound the founding story names (*Giving Is a Gift Too*; the founder's own account is in the essay *The Name Is the First Gift*). The pack's aerial response — a B-Dog™ flying to a bark as an event-gated witness, never on patrol and never as the rescue, its footage encrypted to the holder's pack — is specified in *The Quest That Mints Nothing* (claim 8). The companion family itself — a protector and a courier, a cat that gives without soliciting and a dog that guards without surveilling — belongs to the production arm's service-robot roster, where the animal grammar is a deliberate specification device: the robot's social contract, made legible to a child and enforced by its archetype. In the mate-selection deployment (*Aura-Gated Anonymous Mate-Selection*), the companion ships as the safety layer of first meetings — the population for whom the protector-dog is already a lived institution.

Miss AquariusSM is named throughout as the architecture's eventual autonomous operator; she does not yet exist as such, and every capability attributed to her here is design intent inherited as constitution, not a description of a running system. The alarm grammars specified here require no AI beyond on-device inference; the companion's warmth is a register, not a claim.

14 • Selected lineage

Check-in and journey safety: Apple Check In (2023; Apple Platform Security guide, *Check In security*); the Companion app (University of Michigan students, 2015); Life360 (product tiers; Driver Protect, 2016; Jon Keegan and Alfred Ng, "*The Popular Family Safety App Life360 Is Selling Precise Location Data on Its Tens of Millions of Users*," The Markup, 6 December 2021); Kitestring (2014); Noonlight; bSafe; Garmin inReach; Apple Maps automatic ETA sharing (2019); Google Car Crash Detection (Pixel, 2019); Android Emergency SOS and Emergency Location Service; Apple Crash/Fall Detection. Dead-man and heartbeat: railway dead-man controls; industrial watchdog timers; distributed-systems failure detectors; lone-worker monitoring timers (e.g., SoloProtect); alarm-system duress PINs. Coercive control: Freed et al., CHI 2018 and successors; Chatterjee et al., IEEE S&P 2018; Citizen Lab, *The Predator in Your Pocket*, 2019. Location privacy: de Montjoye et al., *Unique in the Crowd*, 2013; Golle & Partridge, 2009; GDPR Art. 5(1)(c); on-device significant-location processing (Apple, Google). Pre-consent and release-on-event: Google Inactive Account Manager (2013); advance directives and Ulysses contracts (bioethics literature); sealed testamentary instruments. Community response: PulsePoint; Good-Samaritan response networks; critiques of open neighborhood-watch platforms. Anonymous giving: Maimonides, *Mishneh Torah*, Gifts to the Poor 10:7–14; commercial anonymous gifting practice. Corpus companions: *The Thank-All-Nearby Primitive*; *Verified-Human Anonymous Local Gratitude Transfer*; *The Time-Locked Gift Tag*; *Proof of Coordinate*; *B-PoHSM as Humanity Layer*; *The Gift Operation*; *The Unpaid Relay*; *The Quest That Mints Nothing*.

15 • Conclusion

A safety product earns trust by what it can see; it keeps trust by what it refuses to remember. The mechanisms specified here — a companion that notices and asks, a pack for whom silence is the alarm, a cancellation that protects even under a watcher's eyes, strangers verified before they are called, a directive written in daylight that opens one bounded window in the dark, a track that dies on schedule while a diary its owner alone can read and tear survives, and a courier that carries thanks home to a giver with no name — are one design: protection bound to the present, the past bound to its subject, and an institution bound, permanently and in public, to hold neither.

May the bark never be needed. That is what it is for.

Drafted in the HeartBank® research corpus, 4 July 2026. Co-authored with Miss AquariusSM, the architecture's named AI collaborator, per the corpus's standing disclosure: the mechanism synthesis, prior-art survey, and adversarial analysis are genuine collaboration; final editorial control and responsibility for every claim rest with the human author. Dedicated to the public domain under CC0 1.0. Marks including HeartBank®, B-SafetySM, B-SidekickSM, B-PackSM, B-Dog[™], B-QuestSM, B-Cat[™], B-HauntSM, B-DatingSM, B-Yearbook, 3B-Diamond[™], B-Emblem[™], Proof of HumanitySM, PoHSM, Proof of CoordinateSM, PoCSM, and Miss AquariusSM are reserved; the mechanisms are not.